# SpotCast

Synopsis for the app project

SWMAD-01 Mobile Application Development

Author: Nuno Alves Gonçalves, study number 202600724 (the only member of the group)

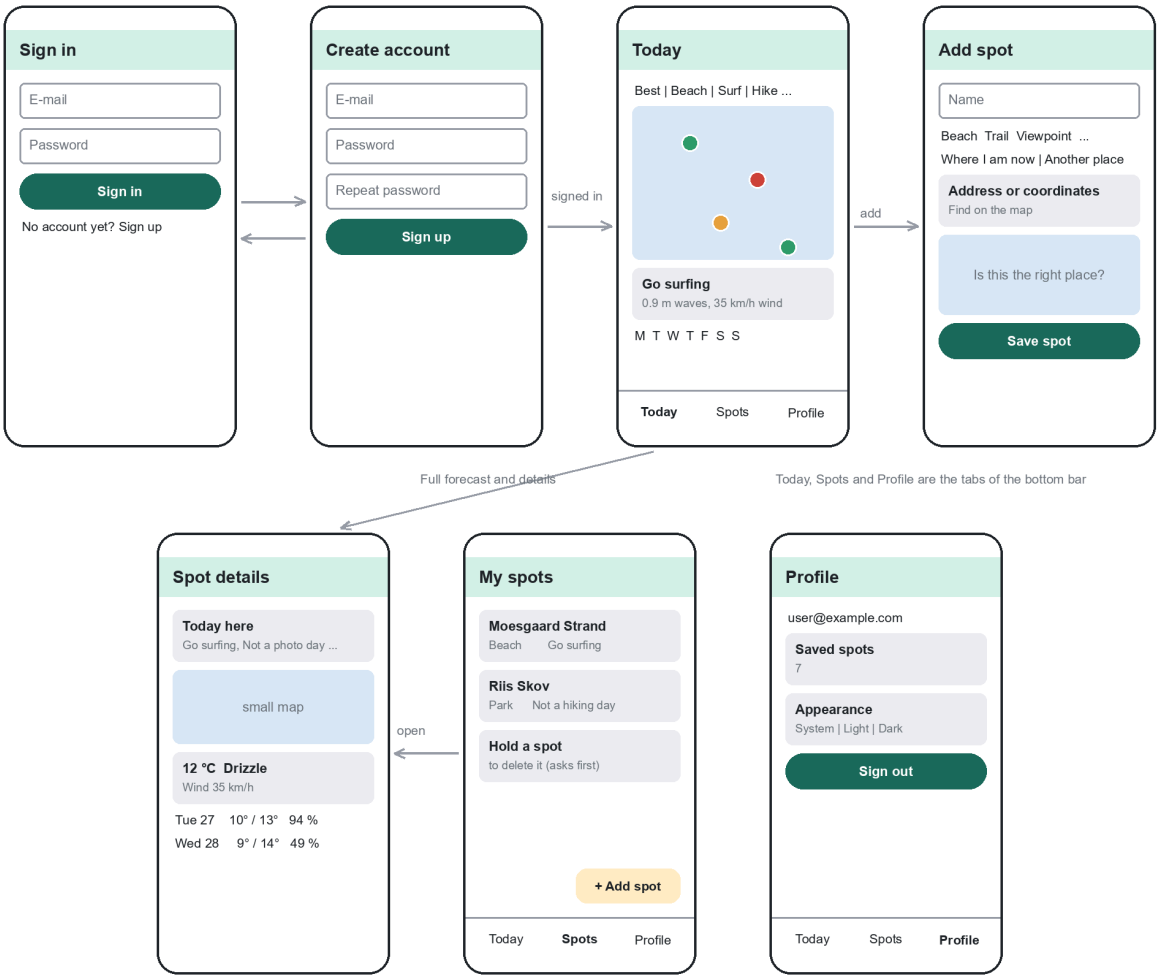
## App vision

People who spend time outdoors keep a mental list of places: a beach that gets waves, a forest for a walk, a pier with a good sunset. A weather app gives them numbers for the nearest town, but not the answer they need: is today a good day to do this, here?

SpotCast answers Go, Maybe or Skip for every saved spot and five activities: a beach day, surfing, hiking, photos and a picnic. The app opens on a map where each pin has the colour of today's answer. Tapping a pin explains the answer, suggests a better activity for that spot, and shows the best day of the week. Spots are saved where the user stands, or anywhere by address, and stay in the user's account.

## Screens

The app has five screens besides sign in and sign up. The layout below shows all seven.




- **Today**: the map with a pin per spot coloured by verdict, an activity bar, and a card with the full report for the selected spot.
- **Spots**: the list of spots with today's best activity; holding a spot deletes it after a confirmation.
- **Add spot**: name, category and the position, from GPS or from an address or coordinates that the user confirms on a map.
- **Spot details**: the verdict for all five activities, a small map, the weather now and the next

seven days.

- **Profile**: the account, the spots per category, the light, dark or system theme, and sign out.

## User stories

- As someone with a free afternoon, I want to see which of my spots are worth it today, so that I do not have to read seven forecasts.
- As a photographer who also surfs, I want to pick an activity and be told when another one suits a spot better.
- As a user, I want to know why a spot is a Skip and which day this week is better.
- As a hiker, I want to save the place where I stand with one button, or a place I have not been to by its address.
- As a user, I want to delete a spot without risking deleting one by accident.
- As a user with a phone and a tablet, I want my spots to follow my account.

## Technologies

- Kotlin, coroutines and Jetpack Compose with Compose Navigation.
- Firebase Authentication with e-mail and password (mandatory).
- Firestore database for the spots of each user, with real-time updates (mandatory).
- Ktor HTTP client calling the Open-Meteo forecast and marine APIs, which are not Firebase services and are called without an SDK (mandatory), and Nominatim for address search.
- GPS through the fused location provider, and a vector map with MapLibre and the free OpenFreeMap tiles (the additional technology).